Webservices – Browse data Xml

manual, version 5.4

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For general information about our webservices see the Twinfield Webservices Manual

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1. Browse data

Twinfield has several functions available to get financial data out of Twinfield, e.g. overview of the general ledger, purchase invoices and time & expenses overviews.

Browse data can be divided into several area's

1. Read the browse definition
2. List of the available browse fields
3. Request of financial transactions
4. Response of the request
5. Budget vs Actuals listing
6. Twinfield Analysis cubes

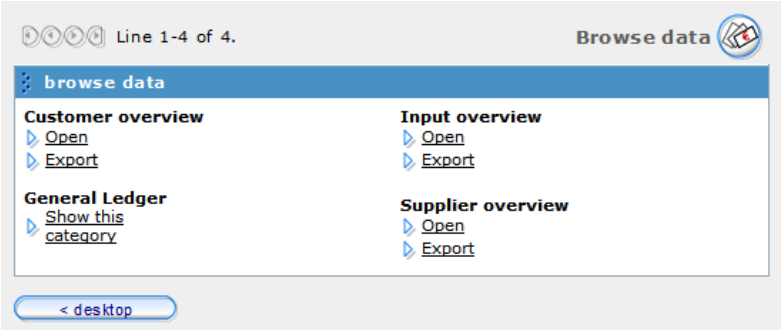
When developing a query to extract data from Twinfield you need to follow these steps:

1. Determine in Twinfield which browse definition you want to use.
2. Read the browse definition
3. Use the columns node in the request from the browse definition.
4. Fill in the selection criteria.
5. Send the (browse) request to Twinfield.

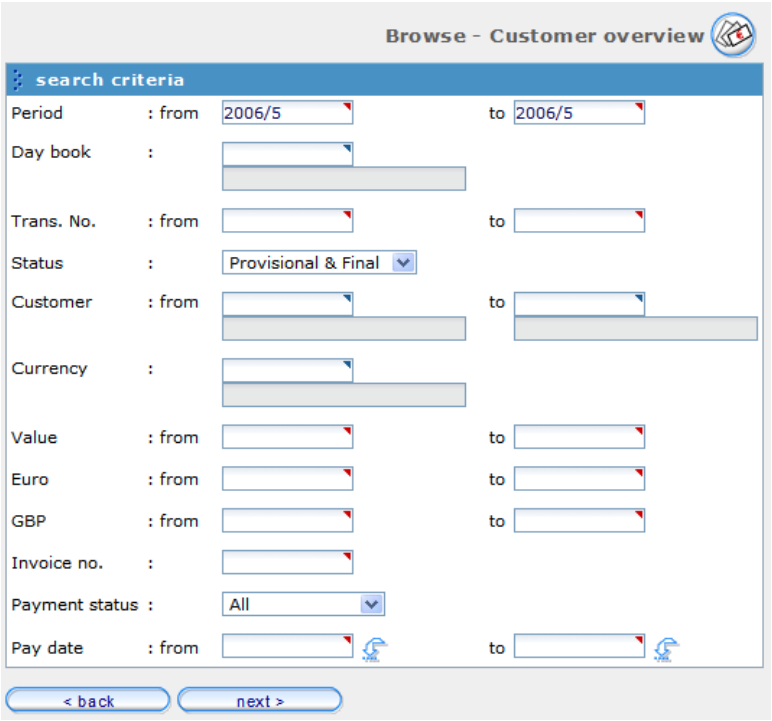
1.1 Determine in Twinfield which browse definition you want to use

Menu path: Accounting>Financials>Browse data

The list of available browse definitions can be found in chapter 2 Browse codes.



Customer overview / Selection criteria



Result screen



1.2 Read the browse definition

Twinfield has a XML message available to read which fields and selection criteria are available in a browse screen.

XML request example

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<read>
  <type>browse</type>
  <office>777</office>
  <code>010</code>
</read>
```

XML response example

```
<browse result="1">
  <office name="Webshop JJLammers.com" shortname="777">777</office>
  <code>010</code>
  <name>Unmatched suspense accounts</name>
  <visible>true</visible>
  <columns code="010">
    <column id="1">
      <field>fin.trs.head.yearperiod</field>
      <label>Periode</label>
      <visible>true</visible>
      <ask>true</ask>
      <operator>between</operator>
      <from>2006/8</from>
      <to>2006/12</to>
      <finderparam></finderparam>
    </column>
    <column id="…"> … </column>
  </columns>
</browse>
```

1.3 List of available browse fields

To get an overview of all available browse fields and selection criteria the following XML message can be used.

XML request example

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<list>
  <type>browsefields</type>
</list>
```

XML response example

```
<browsefields>
  <browsefield>
    <code>fin.rmd.detail.remindcount</code>
    <datatype>long</datatype>
    <finder></finder>
    <dropdown/>
    <canorder>true</canorder>
  </browsefield>
  <browsefield> … </browsefield>
  <browsefield>
    <code>fin.trs.head.status</code>
    <datatype>special</datatype>
    <finder></finder>
    <dropdown>
      <option name="Draft">draft</option>
      <option name="Provisional">temporary</option>
      <option name="Final">final</option>
      <option name="Deleted">deleted</option>
      <option name="In use">inuse</option>
      <option name="Provisional &amp;amp; Final">normal</option>
    </dropdown>
    <canorder>true</canorder>
  </browsefield>
</browsefields>
```

On several browse fields are dropdown values available, the “dropdown” value must be used when entering a selection criteria. The name attribute holds the translation of the “dropdown” label.

1.4 Structure XML request of browse data

Tag	Attributes	Description	Occur
<columns>	code optimize ouputformat	Browse code (see chapter 2 Browse codes). Optimize the XML output (bandwidth) Format of the output XML	1
<sort>			
<field>			
@order			
</field>			
</sort>			
<column>			1-∞
<...></...>			
</column>			
</ columns>			

Attributes

Attribute	Description
optimize	Enables or disables optimization of the output. Possible values are "true" and "false". "false" is the default value. When set to true, the "field", "hideforuser" and "type" attributes are removed from all <td> elements except those in the first <tr> element.
outputformat	Determ the output format of the data. Possible values are "XML" and "CSVEXCEL", "XML" is the default value. Output format XML creates a series of <tr> and <td> elements to store the data. Output format CSVEXCEL creates a single <csv> element containing CSV data optimized for use in MS Excel. The field separator is tab character.

Possible fields (column)

Tag	Parent node	Description	Type
columns	-	root node	-
column	columns	node	-
Field	column	Fieldname	VARCHAR(40)
Label	column	Column header description	VARCHAR(255)
Visible	column	Show the column to the user; values true, false = default	BOOLEAN, optional
Ask	column	Can a selection criteria been entered; Read only, values true, false = default	BOOLEAN, optional
operator	column	List of operators; possible values - none - between - equal - lessthan -	ENUMERATION
From	column	Value from; when the operator is “none” the value will be ignored	VARCHAR(40)

To	column	Value to; only be filled in when the operator is “between”	VARCHAR(40)
Finderparam	column	Which finder must be presented (only available in Twinfield)	VARCHAR(40)

When the operator is set to “none” the “from” and “to” values are ignored. When the “ask” element is “true” a selection criteria can be entered.

Request XML example

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<columns code="100">
  <column id="1">
    <field>fin.trs.line.dim2</field>
    <operator>between</operator>
    <from>1001</from> <!-- Customer -->
    <to/>
  </column>
  <column id="2">
    <field>fin.trs.line.openbasevaluesigned</field>
    <label>Outstanding</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
  <column id="3">
    <field>fin.trs.line.invnumber</field>
    <label>Invoice Number</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
  <column id="4">
    <field>fin.trs.line.matchstatus</field>
    <operator>equal</operator>
    <from>matched</from> <!-- matchstatus -->
  </column>
  <column id="5">
    <field>fin.trs.head.code</field>
    <operator>equal</operator>
    <from>VRK</from> <!-- Journal -->
  </column>
  <column id="6">
    <field>fin.trs.line.matchdate</field>
    <label>Pay date</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
</columns>
```

Response XML example

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<browse first="1" last="3" total="3" benchmark="pre:0.00,db:0.17,post:0.02" result="1">
 <th>
  <td label="Period" hideforuser="false"
    type="String">fin.trs.head.yearperiod</td>
  <td label="General ledger" hideforuser="false"
    type="String">fin.trs.line.dim1</td>
  <td> … </td>
  <td label="Outstanding" hideforuser="false"
    type="Value">fin.trs.line.openbasevaluesigned</td>
  <td label="Payment status" hideforuser="false"
    type="String">fin.trs.line.matchstatus</td>
 </th>
 <tr>
  <td field="fin.trs.head.yearperiod"
    hideforuser="false" type="String">2006/08</td>
  <td field="fin.trs.line.dim1"
    hideforuser="false" type="String">1300</td>
  <td field="fin.trs.line.dim1name"
    hideforuser="false" type="String">Debiteuren</td>
  <td field="fin.trs.head.code"
    hideforuser="false" type="String">VRK</td>
  <td field="fin.trs.head.shortname"
    hideforuser="false" type="String">Verkoop</td>
  <td field="fin.trs.head.number"
    hideforuser="false" type="Decimal">200600107</td>
  <td field="fin.trs.head.curcode"
    hideforuser="false" type="String">EUR</td>
  <td field="fin.trs.line.valuesigned"
    hideforuser="false" type="Value">60.69</td>
  <td field="fin.trs.line.basevaluesigned"
    hideforuser="false" type="Value">60.69</td>
  <td field="fin.trs.line.openbasevaluesigned"
    hideforuser="false" type="Value">60.69</td>
  <td field="fin.trs.line.matchstatus"
    hideforuser="false" type="String"
    name="Available">available</td>
  <key hideforuser="true">
   <office>777</office>
   <code>VRK</code>
   <number>200600107</number>
   <line>1</line>
  </key>
 </tr>
 <tr> … </tr>
</browse>
```

1.5 Budget vs Actuals listing

Description of the budget vs actuals list elements

Tag	Parent node	Description	Type definition
list	-	root node	-
type	list	List type, fixed value: “budgets”	FIXED VALUE
year	list	Fiscal year	INTEGER
periodfrom	list	Fiscal period from	INTEGER
periodto	list	Fiscal period to	INTEGER
budget	list	Budget code	CODE
dim1from	list	GL Account code from	CODE
dim1to	list	GL Account code to	CODE
status	list	Transaction status.	ENUMERATION: - final - concept - both

Example

This example returns a list dimensions with the group hierarchy of the posted actuals compared to the available budgets. If there is no budget available the budget values are ‘0.00’.

Request XML

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<list>
  <type>budgets</type>
  <office>0000</office>
  <year>2007</year>
  <periodfrom>0</periodfrom>
  <periodto>12</periodto>
  <budget>0000</budget>
  <budget>001</budget>
  <dim1from>0000</dim1from>
  <dim1to>9999</dim1to>
  <status>both</status>
</list>
```

Response XML

```
<budgets
        office="0000" officename="Webshop JJLammers.com" year="2007" result="1">
 <budget>
  <period>3</period>
  <dim1 type="BAS" name="Debiteuren">1300</dim1>
 <groups>
  <group id="1" name="Vorderingen">GLB40</group>
  <group id="2" name="Debiteuren">SLB40.10</group>
  <group id="3" name="" />
  </groups>
  <current>
   <debit>578.00</debit>
   <credit>0.00</credit>
   <balance>578.00</balance>
  </current>
  <budget>
   <debit>200.00</debit>
   <credit>0.00</credit>
   <balance>200.00</balance>
  </budget>
  <balance>-378.00</balance>
 </budget>
 <budget>…</budget>
</budgets>
```

1.6 Twinfield Analysis cubes

Twinfield build an Excel Addin to retrieve financial data into an Excel spreadsheet. With this function the users are able to build their own reports and other analysis based on the entered financial data in Twinfield. This function also uses the Webservices functions to request and retrieve the data in Excel.

The Twinfield Analysis flow is as follows:

1. Get a list of available cubes
2. Select and read the corresponding browse code, the same as 'normal' browse.
3. Select the offices to retrieve data from.
4. Fillin the selection criteria, the same as 'normal' browse.
5. Run the browse request per office, the same as 'normal' browse.

List of available cubes

Description of the analysis read elements.

Tag	Parent node	Description	Type definition
Read	-	root node	-
Type	read	Type of read, fixed to 'activeanalysis'	FIXED VALUE

Request XML

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<read>
  <type>activeanalysis</type>
</read>
```

Response description of the analysis elements.

Tag	Parent node	Description	Type definition
activeanalyses	-	root node	-
activeanalysis	activeanalyses		
Code	activeanalyses	Cube code	CODE
Name	activeanalyses	Name	VARCHAR(40)
shortname	activeanalyses		VARCHAR(20)
browse	browse		
Office	browse	Office code, when the office code returns '*' the cube is available in all offices	CODE
Code	browse		

List of available offices

Description of the offices list elements

The complete description and definition of the result XML is described in the document MAN-VALIDATIONS.

Tag	Parent node	Description	Type definition
List	-	root node	-
Type	list	List type, fixed value: “offices”	FIXED VALUE

Request XML

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<list>
  <type>offices</type>
</list>
```

2. Browse codes

Each browse screen has its own definition, the browse codes are listed below per Twinfield function.

Browse data

Code	Description
000	General Ledger overview
010	Unmatched lines
020	Input overview
100	customer overview
163	Open docket invoices
200	Supplier overview
300	Project overview
301	Assets overview
400	Cash overview
410	Bank overview
900	Cost center overview

Twinfield Analysis

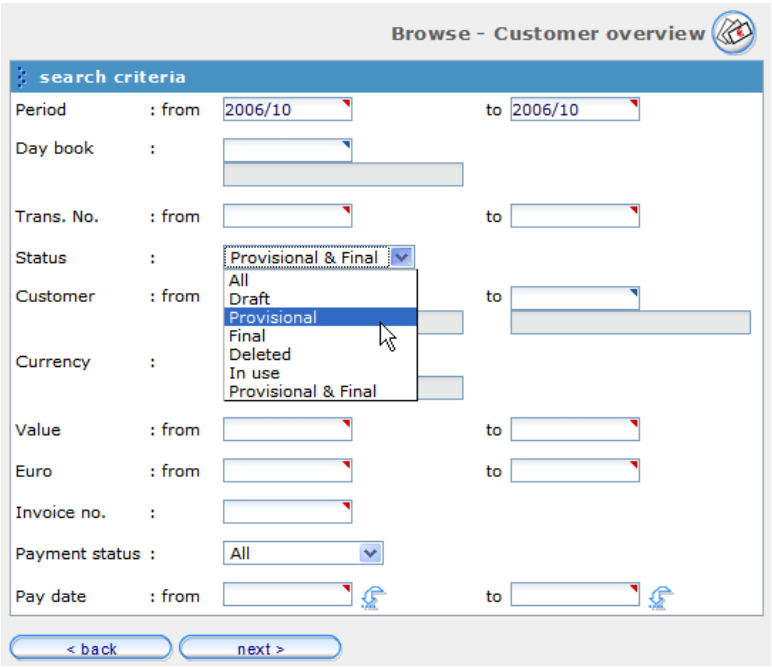
To get a complete list of all available cubes, see also Chapter 1.6 Twinfield Analysis cubes.

Code	Description
030_1	Ledger (details)
031	Ledger (intercompany)
040_1	Annual Report (totals)
050_1	Annual Report (YTD)
060	Annual Report (totals multicurrency)
130_1	Customers
164	Credit Management
230_1	Suppliers
302_1	Fixed Assets
610_1	Time & Expenses (Totals)
620	Project overview (Multicurrency)
650_1	Time & Expenses (Details)
651_1	Time & Expenses (Totals per week)
652_1	Time & Expenses (Totals per month)
660_1	Time & Expenses (Billing details)
661_1	Time & Expenses (Billing per week)
662_1	Time & Expenses (Billing per month)

Example – Customer overview

Menu path: Accounting>Financials>Browse data

Select the option Customer overview. Based on the browse code (100 – customer overview) Twinfield generates the search criteria screen as shown below. Fill in the search criteria and press on the next button to retrieve the requested data. This example we will only search for the paid invoices of customer 1000. The result set will only show the outstanding payment (in case of a partial payment), payment date and the invoice number.



With the Webservices interface you first need to read the browse definition (as described in chapter 2.3) and read the browse fields (as described in chapter 2.4).

XML read browse definition

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<read>
  <type>browse</type>
  <office>0000</office>
  <code>100</code>
</read>
```

To show the search criteria, filter on the nodes where the element "<ask>" is true. To show the dropdown values match the "<field>" element in the browse definition with the field element in the browsefields response.

XML request the payments of customer 1001

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<columns code="100">
  <column id="1">
    <field>fin.trs.line.dim2</field>
    <operator>between</operator>
    <from>1001</from> <!-- Customer -->
    <to/>
  </column>
  <column id="2">
    <field>fin.trs.line.openbasevaluesigned</field>
    <label>Outstanding</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
  <column id="3">
    <field>fin.trs.line.invnumber</field>
    <label>Invoice Number</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
  <column id="4">
    <field>fin.trs.line.matchstatus</field>
    <label>Matched</label>
    <operator>equal</operator>
    <from>matched</from> <!-- matchstatus -->
  </column>
  <column id="5">
    <field>fin.trs.head.code</field>
    <operator>equal</operator>
    <from>VRK</from> <!-- Journal -->
  </column>
  <column id="6">
    <field>fin.trs.line.matchdate</field>
    <label>Pay date</label>
    <visible>true</visible>
    <operator>none</operator>
  </column>
</columns>
```

Fill in the search criteria in the “<from>” and/or “<to>” element.

XML response

This XML response shows the detailed information per field. By using the optimize and/or outputformat attributes in the request XML you can limit the details.

```
<browse first="1" last="2" total="2" benchmark="pre:0.00,db:0.05,post:0.00" result="1">
 <th>
  <td label="Outstanding"
      hideforuser="false" type="Value">fin.trs.line.openbasevaluesigned</td>
  <td label="Invoice Number"
      hideforuser="false" type="String">fin.trs.line.invnumber</td>
  <td label="Matched"
      hideforuser="false" type="String">fin.trs.line.matchstatus</td>
  <td label=""
      hideforuser="false" type="String">fin.trs.head.code</td>
  <td label="Pay date"
      hideforuser="false" type="Date">fin.trs.line.matchdate</td>
 </th>
 <tr>
  <td field="fin.trs.line.openbasevaluesigned"
      hideforuser="false" type="Value">0.00</td>
  <td field="fin.trs.line.invnumber"
      hideforuser="false" type="String">1</td>
  <td field="fin.trs.line.matchstatus"
      hideforuser="false" type="String" name="Matched">matched</td>
  <td field="fin.trs.head.code"
      hideforuser="false" type="String">VRK</td>
  <td field="fin.trs.line.matchdate"
      hideforuser="false" type="Date" name="05/02/2007">20070205</td>
  <key hideforuser="true">
   <office>0000</office>
   <code>VRK</code>
   <number>200600001</number>
   <line>1</line>
  </key>
 </tr>
 <tr>
  <td field="fin.trs.line.openbasevaluesigned"
      hideforuser="false" type="Value">0.00</td>
  <td field="fin.trs.line.invnumber"
      hideforuser="false" type="String">2004043</td>
  <td field="fin.trs.line.matchstatus"
      hideforuser="false" type="String" name="Matched">matched</td>
  <td field="fin.trs.head.code"
      hideforuser="false" type="String">VRK</td>
  <td field="fin.trs.line.matchdate"
      hideforuser="false" type="Date" name="19/02/2007">20070219</td>
  <key hideforuser="true">
   <office>0000</office>
   <code>VRK</code>
   <number>200600009</number>
   <line>1</line>
  </key>
 </tr>
</browse>
```